



ShopBuddy Assistant

What if your shopping assistant actually knew the products - and could prove it??

Garima Narang

Gurwinder Singh

Jayram Karan

Pavankumar Naik

Instructor: Prof.
Mohammad Fawaz

RESEARCH PROBLEM & OBJECTIVE

- LLMs produce **confident but hallucinated** product facts - no real-data connection
- Assistants **lose context** on follow-up questions.
- No audit trail**- users can't verify where answers come from
- E-commerce decisions require **trustworthy, grounded** product intelligence

Objective: Build a grounded, agentic, production-deployed e-commerce assistant that retrieves *real evidence before every answer*- with full transparency on each response.

RESEARCH QUESTIONS & HYPOTHESIS

Does a retrieval-grounded, MCP-orchestrated assistant produce more accurate, consistent answers than direct LLM generation?

H1 – Accuracy

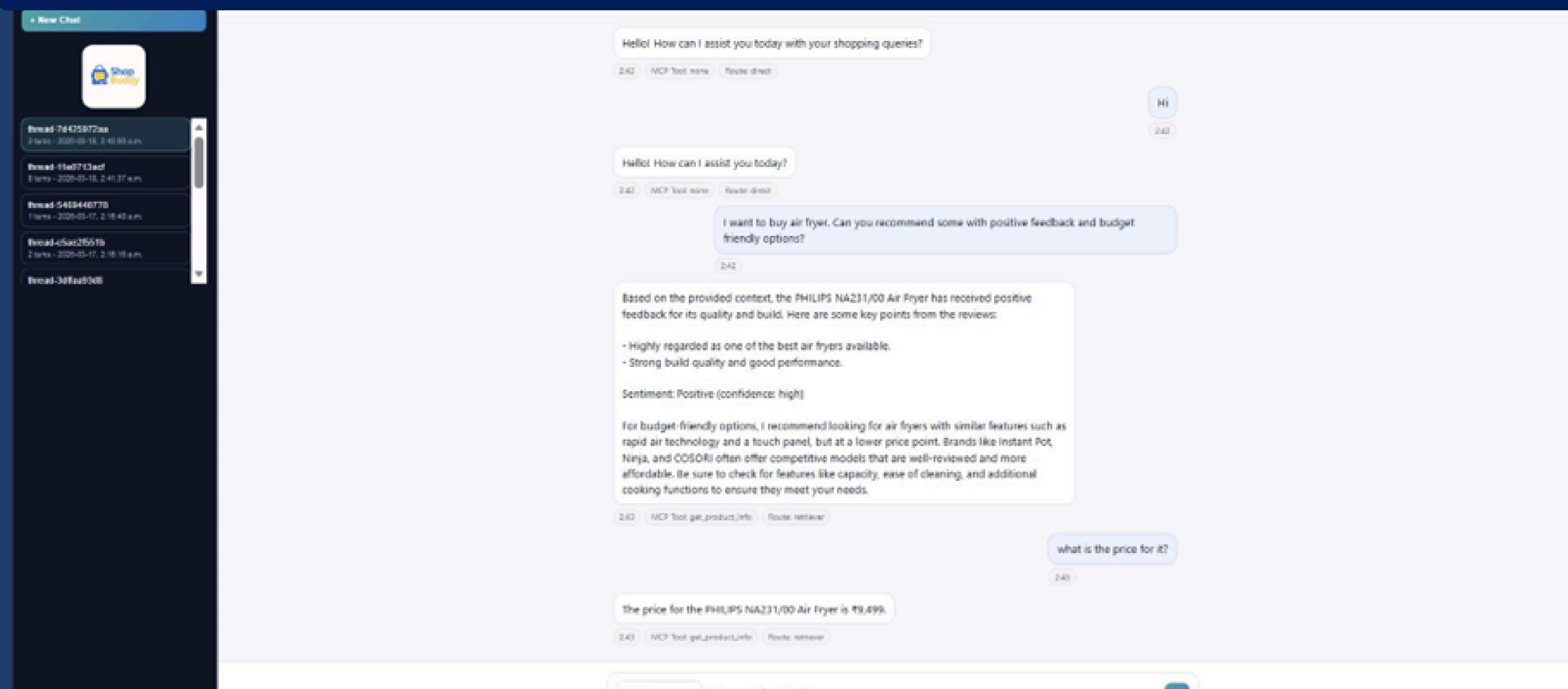
RAG + MCP produces higher factual accuracy and lower hallucination vs. direct generation baseline.

H2 – Consistency

MCP standardized context improves **response consistency & relevancy** vs. Agentic RAG w/o MCP

SEE IT IN ACTION -- LIVE DEMO

Every answer is grounded in **real product data**, and tells you *exactly where it came from*



KEY FINDINGS & STATISTICAL RESULTS

✓ H₁ SUPPORTED

0.25
MCP Accuracy

0.00
Baseline

p=0.0016
Significant ✓

0.57
Cohen's d

95% CI [0.10, 0.40] — does not cross zero. **Medium-large effect size.**

△ H₂ PARTIAL

No-MCP Consistency	1.000
MCP Consistency	0.806
No-MCP Relevancy	0.889
MCP Relevancy	0.839

p=0.089 (n.s.) — **threshold tuning needed.** Honest, actionable finding.

STRATEGIC & PRACTICAL IMPLICATIONS

E-Commerce Teams

Deploy RAG + MCP for product Q&A — near-zero hallucination on specific product facts.

AI Engineers

MCP makes tool backends swappable without workflow rewrites — adopt early for maintainability.

Data & Analytics

Route + RAGAS scores are real-time corpus health signals — monitor continuously.

DevOps / MLOps

100% of outages were config errors. DevOps discipline = equally important as model quality.

LIMITATIONS & FUTURE RESEARCH

△ Limitations

Small Benchmark
n=12, 36 obs — limited power for small effects

Single Source
Flipkart only — India market bias

H₂ Not Supported
Threshold 0.3 needs calibration (p=0.089)

No User Testing
Automated evaluation — no human judges

Future Research

Expand Benchmark
50+ queries, more product categories

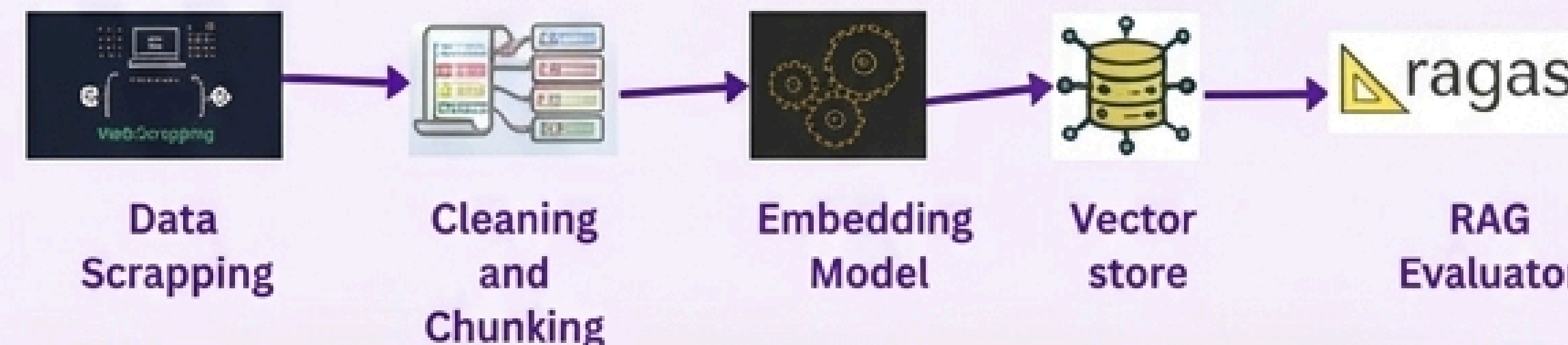
Multi-Platform
Amazon, Best Buy — reduce market bias

Threshold Tuning
Ablation study on RAGAS & MMR lambda_mult

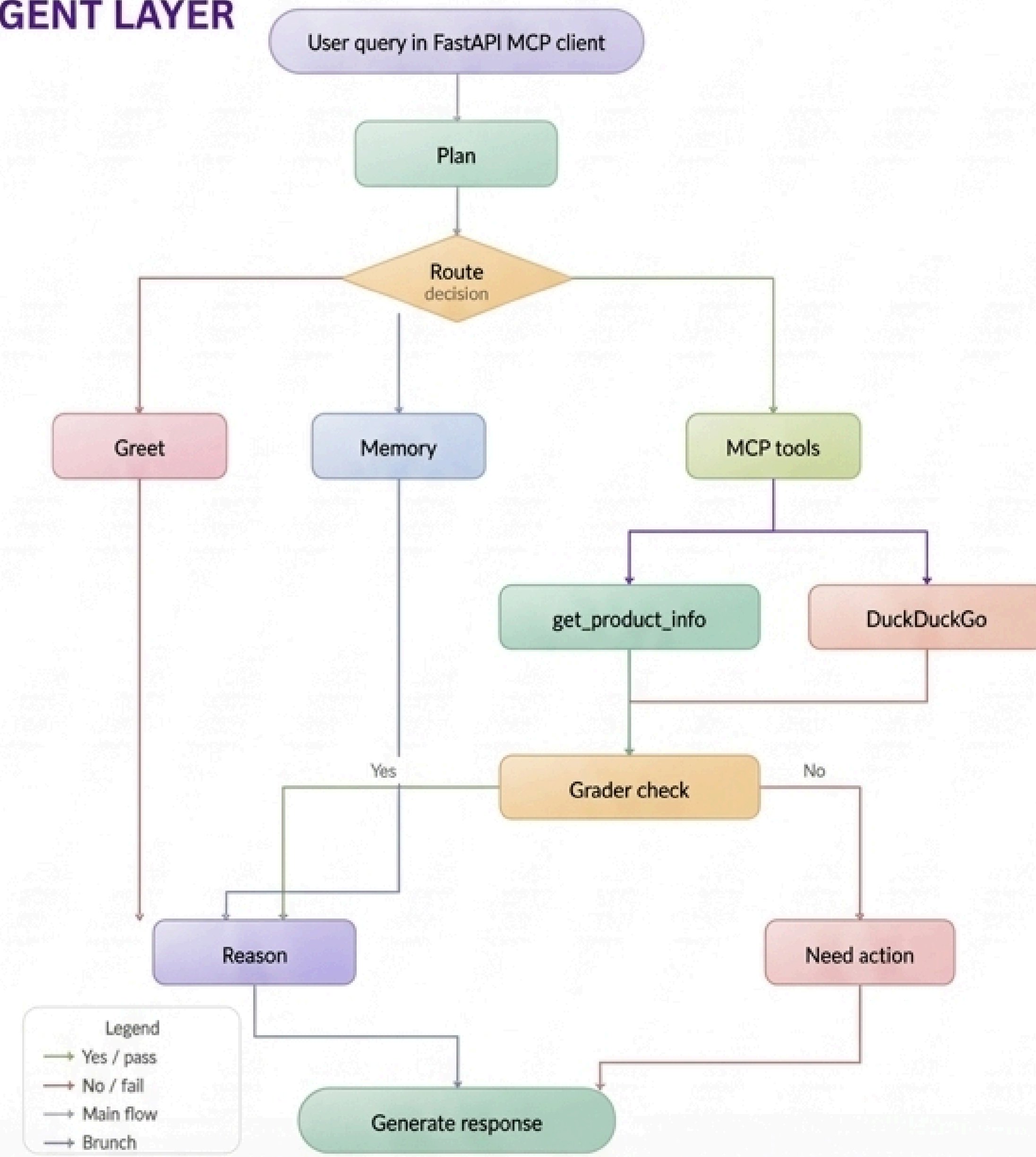
Production Scale
PostgreSQL · OIDC auth · latency observability

ARCHITECTURE

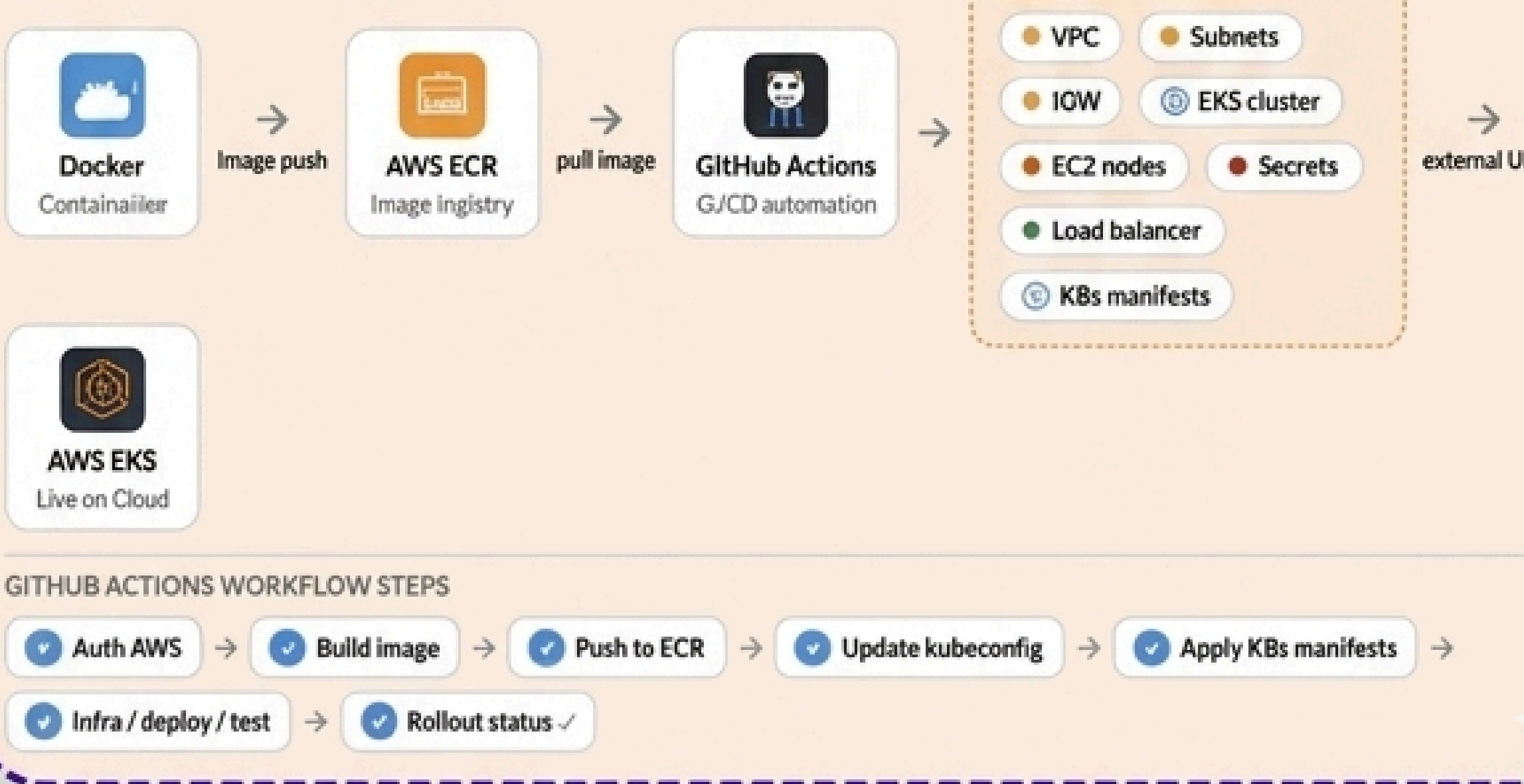
DATA LAYER



AGENT LAYER



AGENT LAYER



SCAN TO TRY SHOPBUDDY LIVE

Feel free to provide feedback @
garimanarang2407gmail.com